

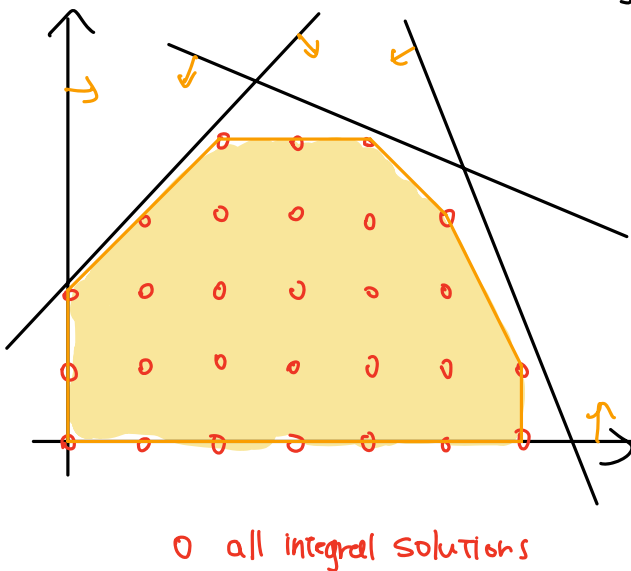
CO 250 Review 10: Integer programs

Integer programs are difficult to solve. There are no known fast algorithms for solving them.

General approach: Solve a related LP (easy), and get good approximations to an optimal integral solution.

Convex hull of integer solutions

Given an IP, we look at all integral feasible solutions. The convex hull of these solutions is the smallest convex set containing all such points.



convex hull of integral solutions.

Meyer's theorem: If all coefficients and constants are rational in the IP, then the convex hull of integral solutions is a polyhedron.

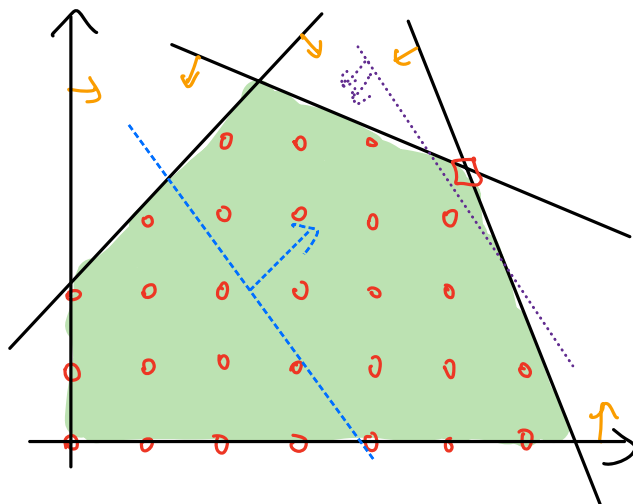
Another result: Optimizing the IP is equivalent to optimizing over the convex hull of integral solutions.

Good: This reduces solving IPs to solving LPs.

Bad: The convex hull is generally very difficult to find (may have exponentially many constraints).

LP relaxation

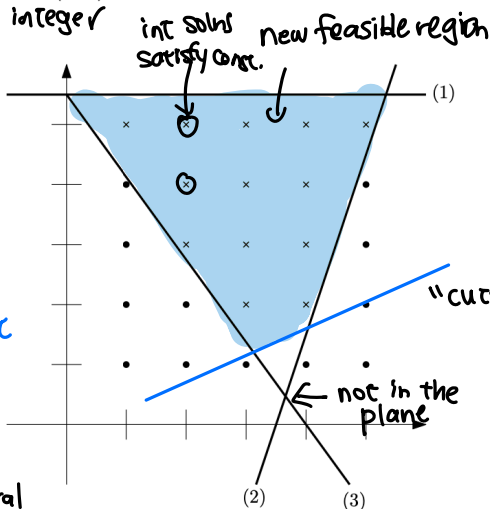
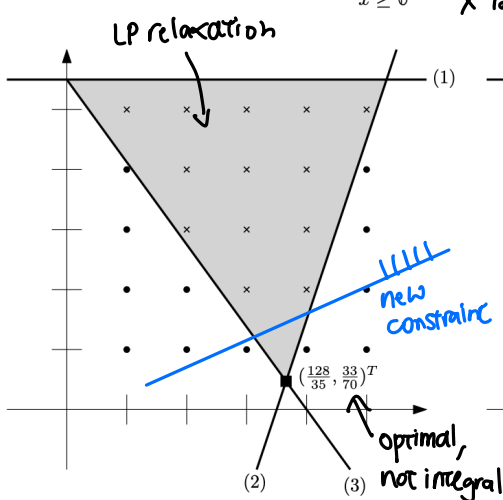
We usually start optimizing an IP by solving its LP relaxation: the LP we get by removing the integrality constraint. Solve it to get an optimal solution $\bar{x}$. If $\bar{x}$ is integral, then it is also optimal for the IP. ($\bar{x}$ is optimal over all points in the LP relaxation, including all integral points.) If $\bar{x}$ is not integral, then we add constraints to exclude $\bar{x}$, and "shrink" the feasible region closer to the convex hull.



- feasible region of the LP relaxation.
- direction of the objective function
- optimal solution to the LP relaxation.
- add a constraint.

Cutting plane method

$$\begin{aligned} \max \quad & (-1 \ -1) x \\ \text{s.t.} \quad & \begin{pmatrix} 0 & 1 \\ 3 & -1 \\ -5.5 & -4 \end{pmatrix} x \leq \begin{pmatrix} 5.5 \\ 10.5 \\ -22 \end{pmatrix} \\ & x \geq 0 \end{aligned} \quad \begin{matrix} (1) \\ (2) \\ (3) \end{matrix}$$



- A cutting plane with respect to a point $\bar{x}$ is a constraint such that
- ① $\bar{x}$ does not satisfy the plane;
 - ② all integral feasible solutions satisfy the plane.

Algebraic method of getting cutting planes: First solve using simplex to get an optimal canonical form.

$$\begin{aligned} \max \quad & (-1, -1, 0, 0, 0)x \\ \text{s.t.} \quad & \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 3 & -1 & 0 & 1 & 0 \\ -5.5 & -4 & 0 & 0 & 1 \end{pmatrix} x = \begin{pmatrix} 5.5 \\ 10.5 \\ -22 \end{pmatrix} \\ & x \geq 0 \end{aligned} \quad \begin{matrix} (1) \\ (2) \\ (3) \end{matrix}$$

$$\begin{aligned} \max \quad & (0, 0, 0, -\frac{3}{35}, -\frac{8}{35})x - \frac{289}{70} \\ \text{s.t.} \quad & \begin{pmatrix} 1 & 0 & 0 & \frac{8}{35} & -\frac{2}{35} \\ 0 & 1 & 0 & -\frac{11}{35} & -\frac{6}{35} \\ 0 & 0 & 1 & \frac{11}{35} & \frac{6}{35} \end{pmatrix} x = \begin{pmatrix} \frac{128}{35} \\ \frac{33}{70} \\ \frac{176}{35} \end{pmatrix} \\ & x \geq 0 \end{aligned}$$

Optimal BFS

$$\bar{x} = \left(\frac{128}{35}, \frac{33}{70}, \frac{176}{35}, 0, 0 \right)^T.$$

not integral.

Take a constraint with a fractional RHS: Pick $x_1 + \frac{8}{35}x_4 - \frac{2}{35}x_5 = \frac{128}{35}$.

Step 1: Take the floor of the coefficients on the LHS. Since $x \geq 0$, the LHS decreases.

$$x_1 + 0 \cdot x_4 - 1 \cdot x_5 \leq \frac{128}{35}.$$

Any feasible solution of the original LP satisfies this.

Note: For integral solutions, the LHS is an integer. So it cannot equal $\frac{128}{35}$.

Step 2: Take the floor of the RHS.

$$x_1 - x_5 \leq 3.$$

Any integral feasible solution satisfies this. But $\bar{x}$ does not satisfy this.

$\Rightarrow$ This is a cutting plane.

Next: Add the cutting plane to the LP (add a slack variable first).

Solve the new LP. (Use dual simplex, CO 370.)

Get another optimal solution. Repeat until we get an integral optimal solution.

Note: This method is guaranteed to produce an integral optimal solution, but it may take a long time to get there.